Respiratory Illnesses

Nasopharyngitis (common cold)

Cause: Viral upper respiratory infection, often rhinovirus.

Findings: Runny/stuffy nose, sneezing, sore throat, cough; fever may occur.

Care: Fluids, rest, saline nose drops and gentle suction. Age-appropriate fever/pain medicine.

No antibiotics for an uncomplicated cold. Avoid OTC cough/cold products in young children.

Sources: CDC pediatric guidance

Allergic Rhinitis

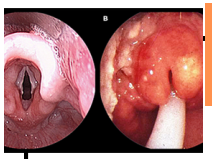
Cause: Allergic response to pollen, dust mites, mold, or animal dander.

Findings: Sneezing, itchy nose/eyes, clear drainage, congestion; fever is not typical.

Care: Reduce allergen exposure, saline rinses, age-appropriate nonsedating antihistamine and/or intranasal corticosteroid as ordered.

Teach regular spray use and aim away from the nasal septum.

Sources: NHS rhinitis



Bacterial Epiglottitis

AIRWAY EMERGENCY: Bacterial swelling above the glottis; Hib is a classic cause.

Findings: Sudden high fever, drooling, painful swallowing, muffled voice, tripod posture; cough often absent.

Priority: Keep calm/upright, NPO; call expert airway team. Do not force throat inspection or use a tongue depressor.

Oxygen if tolerated; controlled airway management and IV antibiotics.

Sources: RCH airway guideline

Pharyngitis (Viral or Bacterial)

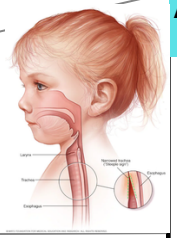
Viral: Often cough, rhinorrhea, or hoarseness. Support fluids and pain relief; no antibiotics.

Group A strep: Fever, sudden sore throat, tender anterior neck nodes; exudate may occur.

Dx: Test when indicated. In symptomatic children age 3+, confirm a negative rapid antigen test with culture.

Tx: Confirmed strep: penicillin/amoxicillin as prescribed; complete the course.

Sources: CDC strep guidance



Acute Laryngotracheobronchitis (Croup)

Cause: Usually viral, often parainfluenza; subglottic inflammation narrows the airway.

Findings: Barking cough, hoarse voice, inspiratory stridor; often worse at night.

Care: Keep child calm with caregiver. Dexamethasone as indicated; nebulized epinephrine for significant obstruction.

Reassess breathing/air entry. Observe after epinephrine per protocol (at least 3 hours in RCH guidance).

Urgent: Stridor at rest, severe retractions, exhaustion, cyanosis, or reduced air entry. No routine antibiotics.

Sources: RCH croup guideline



Tonsillectomy

Indications: Selected recurrent tonsillitis or obstructive sleep-disordered breathing.

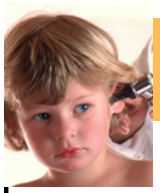
Post-op: Monitor airway, breathing, pain, hydration, and bleeding. Side-lying while drowsy if appropriate.

Hemorrhage clues: Frequent swallowing, fresh blood, bloody emesis, pallor, or tachycardia. Get urgent help; keep NPO if bleeding.

Home care: Fluids and food as tolerated; prescribed acetaminophen/ibuprofen. No aspirin. No codeine/tramadol after tonsil/adenoid surgery in children under 18.

Teach delayed bleeding risk during the first 2 weeks.

Sources: RCH postoperative care, FDA opioid precautions



Otitis Media

Findings: Ear pain, fever, irritability; bulging tympanic membrane with middle-ear fluid.

Care: Analgesia; amoxicillin is often first-line when antibiotics are indicated.

Selected mild cases may use observation for 48-72 hours with reliable follow-up. Reassess persistent pain, fever, or hearing problems.

Sources: CDC pediatric guidance

COVID-19

Cause: SARS-CoV-2 infection.

Findings: Fever, cough, sore throat, fatigue; GI symptoms may occur. Test when clinically indicated.

Care: Fluids, fever/pain relief, and monitor breathing/hydration. Eligible high-risk children may need early antiviral treatment.

Urgent: Respiratory distress, hypoxemia, dehydration, or altered mental status.

MIS-C: Persistent fever with multisystem symptoms 2-6 weeks later needs prompt evaluation. Follow facility infection precautions.

Sources: CDC pediatric COVID-19

Respiratory Illnesses

Respiratory Assessment

History: Onset, cough, fever, exposures, choking, allergies, prematurity, chronic disease, medicines, immunizations, and feeding.

Observe first: Respiratory rate for age, depth/rhythm, retractions, nasal flaring, grunting, posture, ability to speak/cry/feed.

Auscultate: Bilateral air entry, wheeze, crackles, stridor, and diminished/absent sounds.

Monitor: SpO2 trend, pulse, temperature, color, mental status, perfusion, intake/output, and hydration.

Red flags: Apnea, central cyanosis, silent chest, exhaustion, altered consciousness, or bradycardia. Normal SpO2 alone does not rule out distress.

Sources: RCH airway guideline, RCH bronchiolitis

Nursing Interventions for Respiratory Problems

Airway first: Rapid assessment; call for help with severe distress. Position for comfort, often upright; keep caregiver nearby.

Oxygen: Give/titrate to diagnosis-specific orders. Reassess breathing and oxygenation; prepare ventilatory support if deterioration occurs.

Secretions: Gentle nasal suction when obstructed, especially before infant feeds. Avoid routine deep suction.

Hydration: Small frequent feeds if safe. Significant distress may require holding oral feeds and ordered NG/IV fluids.

Other care: Cluster care and allow rest. Give prescribed medicines, reassess response, and follow infection precautions.

Teach warning signs, medication technique, follow-up, and smoke/vape avoidance.

Sources: RCH airway guideline, RCH bronchiolitis

Meds to Treat Respiratory Illnesses

Bronchodilator: Albuterol for bronchospasm/asthma; not routine bronchiolitis treatment.

Steroids: Inhaled for asthma control; systemic for selected attacks. Dexamethasone for croup.

Nebulized epinephrine: Significant croup obstruction; observe for recurrent symptoms.

Antibiotics: For indicated bacterial disease, not uncomplicated viral illness.

Antivirals: Oseltamivir for indicated flu; COVID-19 therapy depends on eligibility.

Symptoms: Age-appropriate acetaminophen/ibuprofen; no aspirin with viral illness. Antihistamines/nasal steroids for allergies.

Verify weight-based orders, allergies, age limits, and response.

Sources: NHLBI asthma treatment, RCH croup guideline, CDC pediatric guidance, CDC flu treatment

Pneumonia

Cause: Viral or bacterial lower respiratory infection.

Findings: Fever, cough, tachypnea, retractions, focal crackles/decreased sounds; chest or abdominal pain.

Care: Assess oxygenation/hydration; oxygen/fluids as needed. Amoxicillin is often first-line for uncomplicated suspected bacterial disease.

Severe distress/hypoxemia needs urgent review. Imaging is not routine for mild uncomplicated illness.

Sources: RCH pneumonia guideline

Influenza A & B

Findings: Abrupt fever, cough, aches, headache, fatigue.

Care: Fluids, fever relief; no aspirin. Start indicated antivirals early, ideally within 48 hours; severe/high-risk cases may benefit later.

Prevention: Annual influenza vaccination from age 6 months.

Sources: CDC flu treatment

Bronchiolitis (RSV)



Cause: Viral small-airway inflammation, commonly RSV; mainly infants/children under 2.

Findings: Cold symptoms followed by cough, tachypnea, wheeze/crackles, retractions, and poor feeding. Young infants may have apnea.

Dx: Usually clinical. No routine chest X-ray or lab testing in typical uncomplicated illness.

Care: Gentle nasal suction for obstruction, hydration/feeding support, and oxygen for persistent hypoxemia per protocol. Monitor apnea and fatigue.

Not routine: Albuterol, corticosteroids, antibiotics, or chest physiotherapy.

Prevention: Hand hygiene and infection precautions. Eligible infants receive protection through maternal RSV vaccination or a long-acting infant antibody.

Sources: RCH bronchiolitis, CDC RSV prevention

Asthma

Cause?

Chronic airway inflammation with variable bronchoconstriction and mucus. Genetic susceptibility plus triggers: allergens, viral illness, exercise, cold air, smoke/vaping.

Sources: NHLBI causes/triggers

Assessment

Recurrent wheeze, cough (often at night), breathlessness, chest tightness, prolonged expiration.

Assess triggers, nighttime symptoms, activity limits, rescue use, prior severe attacks, adherence, and inhaler technique.

Check work of breathing, air entry, respiratory rate, pulse, SpO2, and peak flow if able.

Danger: Silent chest, inability to speak, exhaustion, cyanosis, or altered consciousness.

Sources: NHLBI asthma diagnosis, NHLBI action plan

Dx

History: Variable symptoms and triggers.

Spirometry: Demonstrate airflow limitation and improvement after bronchodilator when child can perform testing (often age 5-6+).

Peak-flow variation may support monitoring. Allergy testing or FeNO may help selected cases.

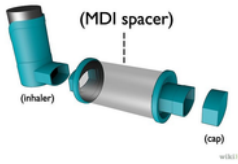
Young children: clinical pattern, response to treatment, and exclusion of other causes.

Sources: NHLBI asthma diagnosis

Treatment/medications

Acute: Prescribed reliever, oxygen for hypoxemia, systemic steroids as ordered. **Long-term:** Age-appropriate inhaled corticosteroid (ICS)-containing plan plus trigger control.

Bronchodilators



Albuterol: Quick relief of bronchospasm.

Ipratropium: Add for severe acute attacks.

LABA: Use only with ICS; never alone for asthma.

Watch for tremor and fast pulse.

Sources: NHLBI asthma treatment

Anti-inflammatory

ICS: Budesonide or fluticasone controls airway inflammation.

Use consistently; rinse mouth afterward. Monitor growth and local thrush.

Systemic steroids: Short course for significant exacerbation as ordered.

Specialist therapy for selected severe asthma.

Sources: NHLBI asthma treatment

Priority Education

Demonstrate MDI/spacer use; have child/caregiver teach it back. Know controller versus rescue medicine and use the written action plan.

Avoid smoke/vape exposure and known triggers. Keep rescue medicine available at home/school. Rinse mouth after ICS.

Track symptoms and peak flow if prescribed. Increasing reliever use or nighttime symptoms need review.

Emergency: Severe breathlessness, blue/gray color, or failure of rescue treatment. Call emergency services; do not delay for a peak-flow reading.

Color-zone percentages refer to the child's personal-best peak flow.

Sources: NHLBI action plan, NHLBI asthma treatment

GREEN: 80-100%
Symptoms controlled
Continue daily plan

YELLOW: 50-79%
Worsening symptoms
Use action-plan reliever

RED: below 50%
Rescue medicine now
Urgent/emergency help

Complications

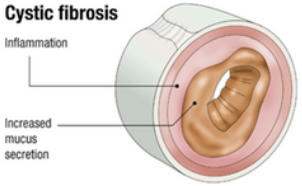
Status asthmaticus, mucus plugging/atelectasis, hypoxemia, and respiratory failure. A silent chest with exhaustion suggests critical obstruction and requires emergency help.

Cystic Fibrosis

Cause?

Autosomal recessive CFTR disorder causes abnormal chloride transport and thick mucus. Two carrier parents: 25% risk each pregnancy.

Sources: CF Foundation overview



Assessment

Salty skin, chronic productive cough, recurrent lung infections, wheezing, and clubbing.

Poor growth despite appetite; bulky greasy stools. Newborns may have meconium ileus.

Track respiratory status, SpO2, sputum, weight/growth, stool pattern, hydration, and salt loss.

Assess treatment burden, adherence, family support, and glucose symptoms.

Sources: CF Foundation overview

Dx

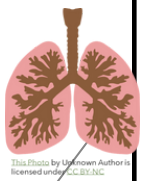
Newborn screen: Elevated IRT, with DNA testing per program; not diagnostic.

Sweat chloride: ≥60 mmol/L supports CF; 30-59 is intermediate and needs evaluation.

Confirm through a CF center with repeat sweat and/or CFTR genetic testing.

Sources: CF Foundation sweat test

Pulmonary

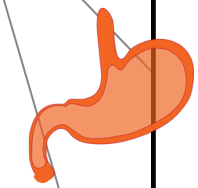


Thick mucus obstructs airways and traps pathogens. Recurrent infection and inflammation cause bronchiectasis.

Chronic cough/sputum, wheeze, crackles, exercise intolerance; later hypoxemia or respiratory failure.

Sources: CF Foundation lung care

GI

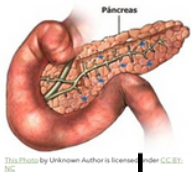


Blocked pancreatic ducts reduce digestive enzymes. Fat/protein malabsorption causes greasy stools and poor growth.

Risks: fat-soluble vitamin deficiency, constipation/distal intestinal obstruction, meconium ileus, and liver disease.

Sources: CF Foundation overview, CF Foundation enzymes

Endocrine



Pancreatic damage reduces insulin production, causing CF-related diabetes (CFRD). Illness may increase insulin resistance.

Look for poor weight gain, increased thirst/urination, fatigue, or declining lung function. CFRD may be asymptomatic.

Sources: CF Foundation diabetes

Treatment/Nursing Considerations



Airway clearance: Daily prescribed techniques/vest and exercise.

Bronchodilator if ordered, hypertonic saline/dornase alfa, and culture-directed antibiotics.

CFTR modulators if eligible. Clean equipment; follow infection precautions.

Sources: CF Foundation mucus therapy, CF Foundation overview



Pancreatic enzymes with meals/snacks when pancreatic-insufficient; do not crush enteric-coated beads.

Individualize calories/protein; give prescribed vitamins A, D, E, K.

Extra salt/fluids as directed, especially with heat/exercise. Track stools/weight.

Sources: CF Foundation enzymes



CFRD: Insulin therapy with CF/endocrine team.

Annual oral glucose tolerance test from age 10; earlier if symptomatic. A1C alone may miss CFRD.

Monitor glucose, adjust insulin, teach hypoglycemia care, and maintain CF nutrition.

Sources: CF Foundation diabetes